Lab Tutorial: Multivariate Linear Regression

Introduction

This lab tutorial covers three approaches to implementing multivariate linear regression for predicting house prices based on size (square feet) and the number of bedrooms. You will explore the following methods:

1. Manual Gradient Descent
2. Normal Equation
3. Built-in Gradient Descent using scikit-learn

Through these exercises, you will gain a deeper understanding of regression techniques and their applications.

Dataset Description

The dataset lab2data.txt contains three columns:

- Size (square feet)
- Number of Bedrooms
- Price (in USD)

Exercise 1: Linear Regression with Gradient Descent

Objective: Implement multivariate linear regression using gradient descent to predict house prices.

Steps:

1. Load the dataset using numpy.
2. Normalize features for better performance.
3. Implement gradient descent to minimize the cost function:
   - Update rule: $\theta_j := \theta_j - \alpha \, \partial J(\theta)/\partial \theta_j$
4. Evaluate the model using MSE.
5. Predict the price for a house with 2500 sq ft and 4 bedrooms.

Exercise 2: Linear Regression with the Normal Equation

Objective: Implement multivariate linear regression using the normal equation to predict house prices.

Steps:

1. Load the dataset using numpy.
2. Add a bias term to the feature matrix.
3. Compute the optimal θ using the formula: $\theta = (X^T X)^{-1} X^T y$.
4. Evaluate the model using MSE.
5. Predict the price for a house with 2500 sq ft and 4 bedrooms.

Exercise 3: Linear Regression with Built-in Gradient Descent

Objective: Use scikit-learn's LinearRegression to implement multivariate linear regression.

Steps:

1. Load the dataset using numpy.
2. Normalize features using StandardScaler from scikit-learn.
3. Train a LinearRegression model.
4. Evaluate the model using MSE.
5. Predict the price for a house with 2500 sq ft and 4 bedrooms.